Tessa Adams
CariSurg Healthcare AI Program
Week 7 Cost Benefit Memo
Date: July 17, 2026

Week 7 Cost-Benefit Memo: Model Selection for Phase 3

Recommendation

I recommend carrying Logistic Regression into Phase 3 because it achieved the strongest overall performance while remaining easier to interpret, maintain, and deploy than the more complex models tested.

Methods

The analysis used 55,121 emergency department encounters and reused the Week 6 train-test split to ensure that all models were evaluated on the same patients. Models used information available at triage, including vital signs and chief-complaint indicators. Post-triage information was excluded to prevent data leakage.

Logistic Regression was compared with Random Forest, Gradient Boosting, and a small multi-layer perceptron (MLP). Models were evaluated using predictive performance, training and inference time, and interpretability.

Inspired by the TriageIntelli study, which found that stacking could improve triage performance, a Logistic Regression–Random Forest ensemble was also tested. It improved ESI 1 recall to 0.69, but its macro-F1 fell to 0.440 and ESI 1 precision was only 0.02, so it was not selected.

Benchmark Results

Model	Accuracy	Macro Precision	Macro Recall	Macro-F1	Training time (minutes)	Inference time per prediction (ms)	Interpretability
Logistic Regression	0.667	0.582	0.463	0.492	0.10	0.00369	High
Tuned Random Forest	0.605	0.449	0.517	0.471	5.35	0.07815	Moderate
Small MLP	0.642	0.518	0.442	0.467	4.38	0.00520	Low
Stacked Ensemble	0.593	0.428	0.614	0.440	13.47	0.05679	Low
Gradient Boosting	0.556	0.423	0.559	0.432	0.22	0.02053	Low to moderate

Untuned Random Forest	0.638	0.468	0.363	0.383	1.08	0.15208	Moderate

Overall, Logistic Regression was the best choice because it had the strongest overall performance and was much simpler and faster to train. Testing more Random Forest settings did not improve its results and took about 34.6 minutes, which further supports choosing Logistic Regression.

Arguments Supporting Logistic Regression

1. It achieved the highest macro-F1 score at 0.492.
2. It is easier to explain and audit than the more complex models.
3. It has a lower training, deployment, and maintenance burden.

Arguments Against Logistic Regression

1. It may miss complex interactions between clinical features.
2. Its overall score may hide weaknesses for ESI 1 and ESI 2 patients.
3. It has not yet been validated across other data splits, time periods, or hospitals.

Risks and Unknowns

ESI Level 1 cases are rare, so performance estimates for this group are based on a small number of patients and may be unstable.

Macro-F1 does not fully reflect the different clinical consequences of undertriage and overtriage. In particular, assigning an ESI 1 or ESI 2 patient to a less urgent level may be more harmful than overtriaging a lower-acuity patient.

The results are based on one dataset and one train-test split. Additional validation is needed across different splits, time periods, patient groups, and clinical settings before deployment.

Final Recommendation

Logistic Regression should proceed as the primary Phase 3 model because it currently provides the best balance of performance, interpretability, computational cost, and deployability.

The tuned Random Forest should remain as a comparison model. Before deployment, both models should undergo critical-class error analysis, repeated validation, fairness assessment, and review of undertriage and overtriage.